Higher National Diploma in Engineering (Civil)
Fourth Year 1st semester Examination- 2019
CE 3222 –Foundation Engineering

Instructions for Candidates:	No. of questions	: 06
Answer any 5 questions.	No. of pages	: 04
Calculators are allowed	Time	: 3 hours

---

01. i) Discuss the steps in planning for sub-soil exploration. (03 marks)

ii) Write down four numbers of information required from site investigation. (04 marks)

iii) What are the data collected in Cone penetration test. (03 marks)

iv) Briefly explain the advantages of Cone penetration test. (04 marks)

v) Briefly explain the following methods of sub-soil exploration.

a.) Vane shear test.

b.) Seismic refraction survey

c.) Electrical resistivity survey (06 marks)

02. i) State three types of shallow foundations and briefly explain them with sketches. (03 marks)

ii) List out the limitations of Terzaghi's bearing capacity equation. (03 marks)

iii) State and explain three types of shear failures in soil. (06 marks)

iv) A strip footing of width of 3m is found at a depth of 2m below the ground surface in soil having a cohesion C = 10 kN/m^2 and friction angle Φ =20°. The water table is at a depth of 4 m below the ground level. Determine the allowable bearing capacity for FOS=3.Use general shear failure theory of Terzaghi.

Dry unit weight of soil =18 kN/m^3 .

Saturated unit weight of soil = 20 kN/m^3 (08 marks)

03. i) Briefly describe the Plate load test procedure with sketches. (03 marks)

ii) Derive the following relationship starting from the definitions.

s.e = w.G_s

Here, s =degree of saturation, e = void ratio, w=water content, G_s= specific gravity of solids (03 marks)

iii) Determine the water content, void ratio, porosity and degree of saturation of following soil core sample.

Weight of soil sample = 1015 g Specific gravity of solids = 2.65

Volume of soil sample = 585 cm^3. Dry weight of soil = 904 g

(06 marks)

iv) Discuss the steps of foundation design. (05 marks)

v) Explain the objectives of following laboratory tests.

a.) Sieve analysis test. c.) Sand cone test

b.) Tri axial test (03 marks)

04. i) Give three examples for internally stabilized earth retaining systems. (03 marks)

ii) Briefly describe the following structures.

a.) Gabion walls. c.) Diapham walls

b.) Sheet piles (06 marks)

iii) A mass concrete retaining wall has to be designed to retain a fill of 8 m. The design parameters of back fill material are mentioned in Figure 04. Determine the lateral earth force and evaluate the stability of the retaining wall against overturning.
Assume the wall and the base are in smooth condition. The ground water table is well below the ground surface. The unit weight of concrete can be taken as 24kN/m^3.

(07+04 marks)

05. i) Briefly explain four situations where the deep foundations are required. (04 marks)

ii) Name two types of piles according to the method of installation and discuss the advantages in each method.

(01+06 marks)

iii) Sub-surface soil profile at a building site is shown in Figure 05. Based on the site investigations and laboratory tests, the relevant soil parameters of each soil layer were found. Assuming that a solid concrete pile with 0.3 m x 0.3 m is driven into 9m depth, calculate skin frictional components in each layer using ∞ and β method.

(09 marks)

06. i) Briefly discuss about three different types of mat foundations with sketches.

(03 marks)

ii) Determine the net ultimate bearing capacity of a mat foundation of 15 m x 10 m on a saturated clay with Cu = 60 kN/m^2 and Φ= 0.It is expected to construct the foundation 2m below the ground level. Assume water table is below the foundation. Following equation with usual notations may be useful in the calculations.

$$q_{(net)} = 5.14\,Cu\left[1+\frac{0.195\,B}{L}\right]\left[1+\frac{0.4\,D_f}{B}\right]$$

(03 marks)

iii) The mat foundation has the dimensions of 30.5 m x 18.3 m. The total dead and live load on the mat is 110 x 10^3 kN. The mat foundation is placed over a saturated clay having a unit weight of 18 kN/m^3 and Cu = 130 kN/m^2 .Given that $D_f = 1.5\ m$.Determine the factor of safety against the bearing capacity failure.

(06 marks)

iv) Briefly explain four situations where caissons are required. (04 marks)

v) List out four types of caissons and briefly explain them. (04 marks)

Table Q2.iv: Bearing capacity factors shallow foundations

Φ degrees	N_c	N_q	N_r
0	5.7	1.0	0.0
5	7.3	1.6	0.14
10	9.6	2.7	1.2
15	12.9	4.4	1.8
20	17.7	7.4	5.0
25	25.1	12.7	9.7
30	37.2	22.5	19.7
35	57.8	41.4	42.4
40	95.7	81.3	100.4
45	172.3	173.3	360.0
50	347.5	415.1	1072.8

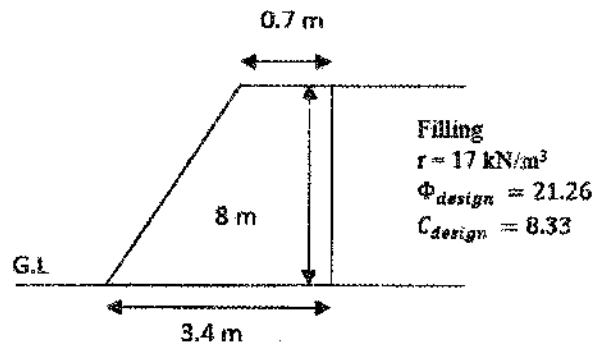

Figure Q4.iii: Mass concrete retaining wall

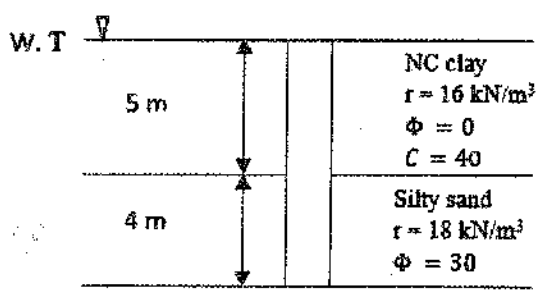

Figure Q5.iii: Sub-surface soil profile for the pile driven